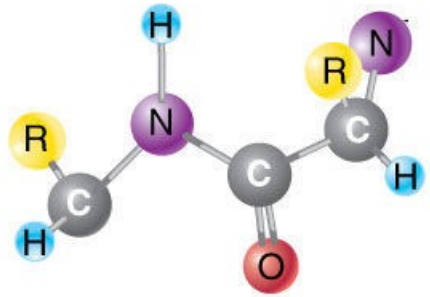
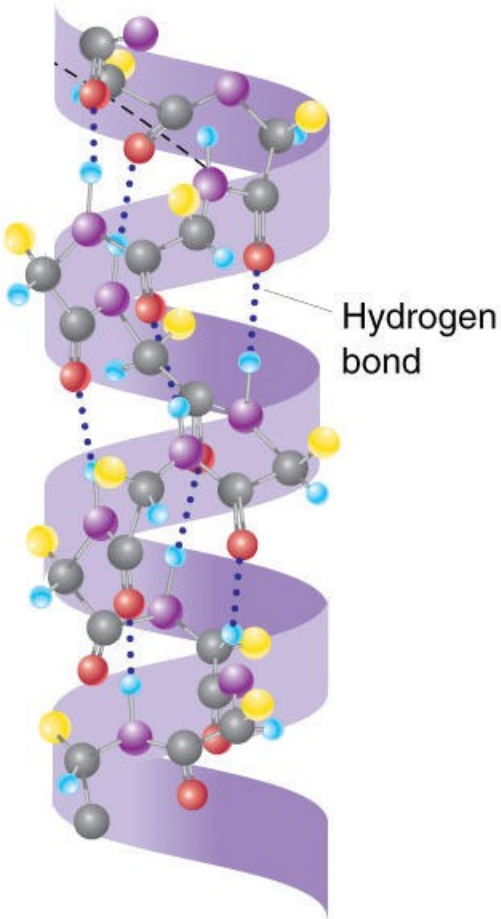
Module 7

Macromolecular Analysis

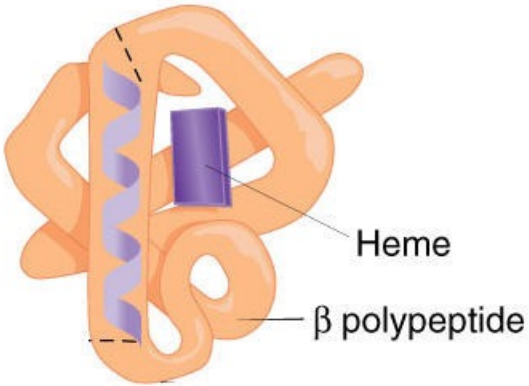
Protein Structure



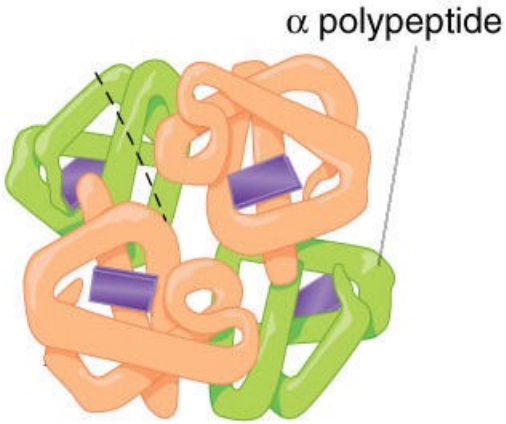
(a) Primary structure



(b) Secondary structure



c) Tertiary structure



(d) Quaternary structure

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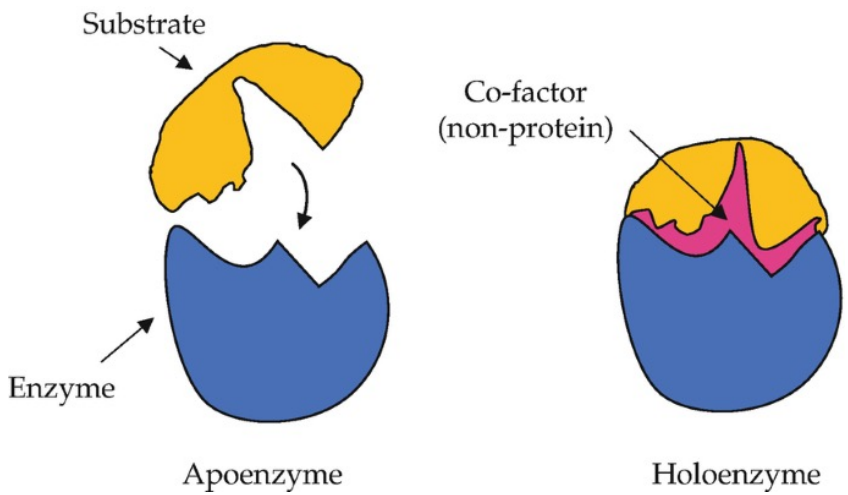
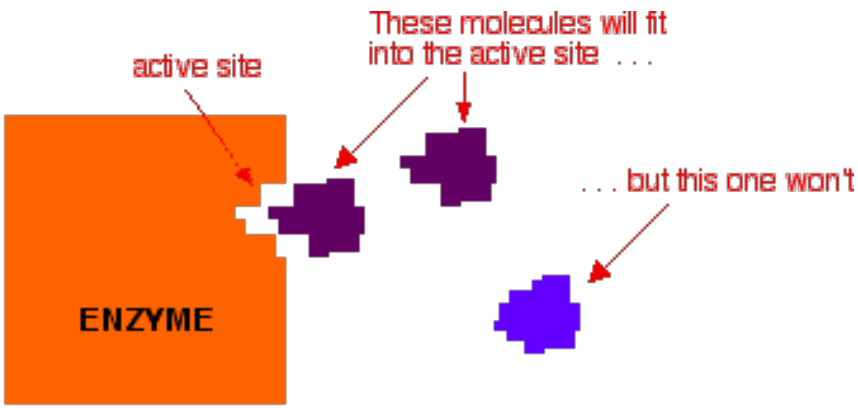
Protein as Enzymes

- Enzymes are specialized proteins that act **as biological catalysts**, **accelerating chemical reactions** within living organisms
- Enzymes speed up chemical reactions by **lowering the activation energy** required for the reaction to occur.
- The **active site** of an enzyme is a specialized region where **substrate binding** and **catalysis** occur
- Many enzymes require **non-protein molecules** for activity. These include **metal ions (cofactors)** and **organic molecules (coenzymes, e.g., NAD^+, FAD)**.
- The behavior of enzymes is often described by **Michaelis-Menten kinetics**, characterized by parameters like $\mathbf{V_{max}}$ (maximum velocity) and $\mathbf{K_m}$ (Michaelis constant), reflecting enzyme efficiency and substrate affinity.
- The catalytic **efficiency** and **specificity** of enzymes are intrinsically linked to their **tertiary** and **quaternary structures**. ***Denaturation can lead to loss of enzymatic activity.***
- Enzymes are involved in virtually all **cellular processes**, including metabolism, DNA replication and repair, signal transduction, and cellular defense mechanisms, highlighting their central role in biology.





Biologically relevant enzymes

DNA Polymerase

- Function: Catalyses the synthesis of DNA from deoxyribonucleotides during replication.
- Biological Relevance: Essential for cell division and genetic inheritance.
- Medical Relevance: Errors in function can lead to mutations.

Pepsin

- Function: A protease that breaks down proteins into peptides in the stomach.
- Biological Relevance: Protein digestion under acidic pH conditions (~pH 2).
- Medical Relevance: Deficiency or dysfunction can lead to impaired digestion and malnutrition.

Hexokinase

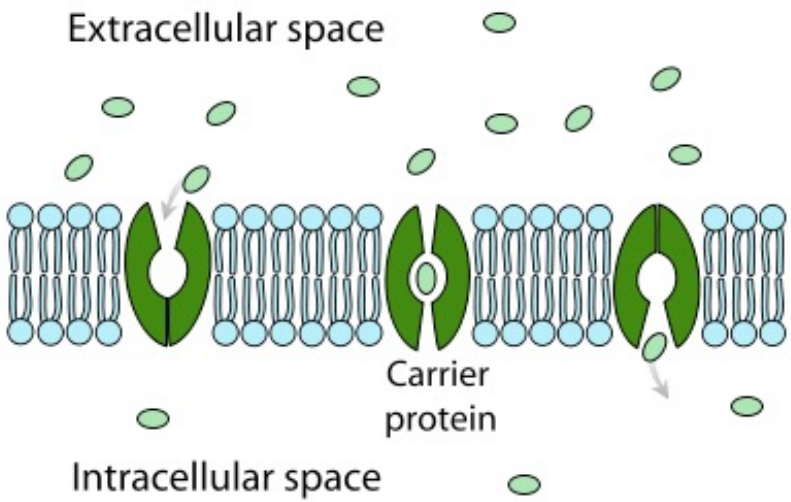
- Function: Catalyzes the phosphorylation of glucose to glucose-6-phosphate, the first step in glycolysis.
- Biological Relevance: Regulates glucose utilization for energy production.
- Medical Relevance: Overexpressed in cancer cells (Warburg effect), making it a potential cancer biomarker

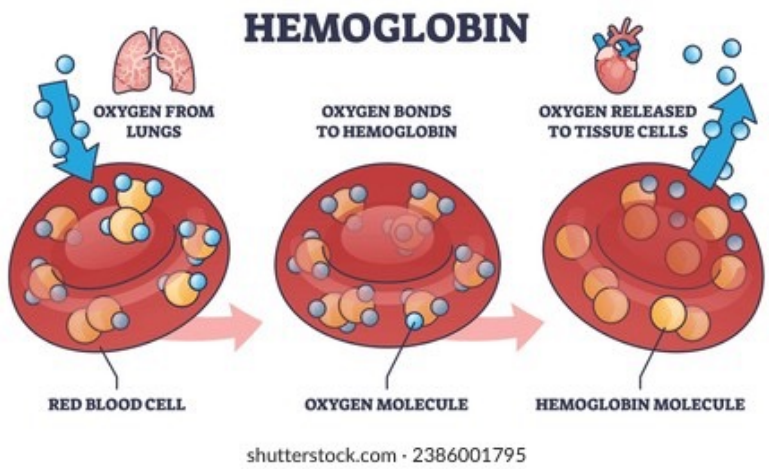
Nitric Oxide Synthase (NOS)

- Function: Synthesizes nitric oxide (NO) from L-arginine.
- Biological Relevance: NO is a signaling molecule involved in vasodilation, immune response, and neurotransmission.Medical Relevance: Dysregulation is associated with cardiovascular diseases, neurodegeneration, and inflammation.

Protein as Transporters

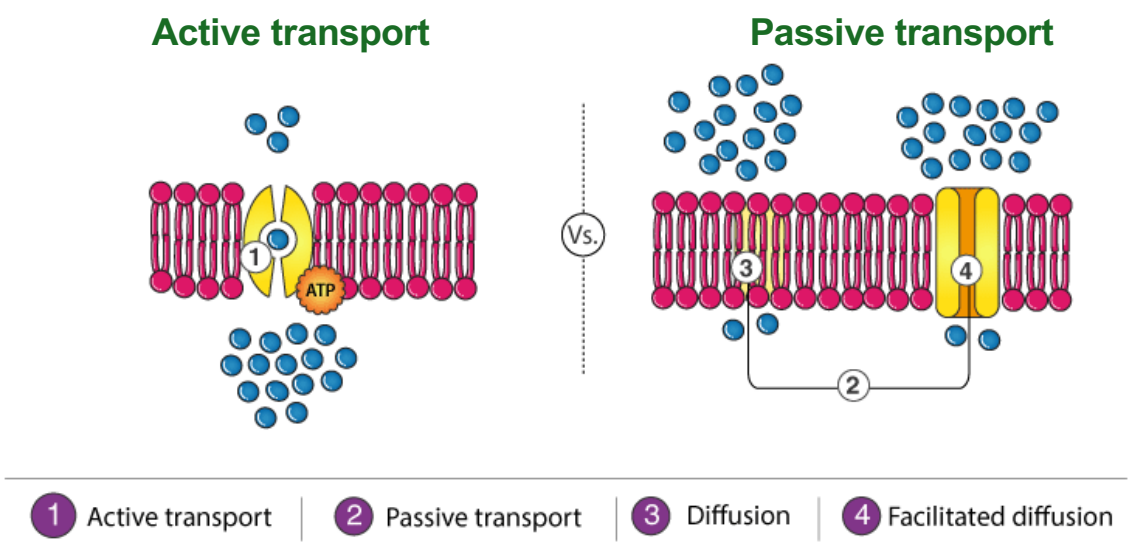
- Transport proteins **enable the movement** of ions, nutrients, gases, and other molecules across cellular membranes, which are otherwise impermeable to many solutes.
- Most transport proteins are **integral membrane proteins** embedded in the lipid bilayer, allowing them to span the membrane and interact with both intracellular and extracellular environments.
- Transport proteins exhibit **high substrate specificity**, often transporting only one type of ion or molecule (e.g., glucose transporters, K^+ channels).
- Transporters affect drug absorption, distribution, and elimination. Many drugs target or depend on transporters.





Types of Transport Mechanisms

- ❑ **Passive (facilitated diffusion):** No energy required; down concentration gradient.
- ❑ **Active transport:** Requires energy (often ATP) to move substances against their gradient.



FEATURE	ACTIVE TRANSPORT	PASSIVE TRANSPORT
Energy Usage	Requires ATP	None required
Concentration Gradient	Moves against gradient	Moves along gradient
Transport Proteins	Necessary (pumps, carriers)	Optional (channels, carriers), or none
Examples of Molecules	Ions (Na+, K+), glucose	Oxygen, CO2, water
Speed	Slower due to energy use	Faster
Regulation	Highly regulated	Less regulated

Mechanism of Passive Transport

No Energy Required

•Moves substances **down** their concentration or electrochemical gradient.
•Does **not require ATP** or other energy input.
•Occurs through **simple diffusion**, **facilitated diffusion**, or **channel proteins**.
•Transport rate depends on the **concentration gradient** and **membrane permeability**.
•Examples:

- Oxygen and CO_2 diffusion
- Water movement via aquaporins
- Ion flow through open channels (e.g., K^+ leak channels)

Mechanism of Active Transport

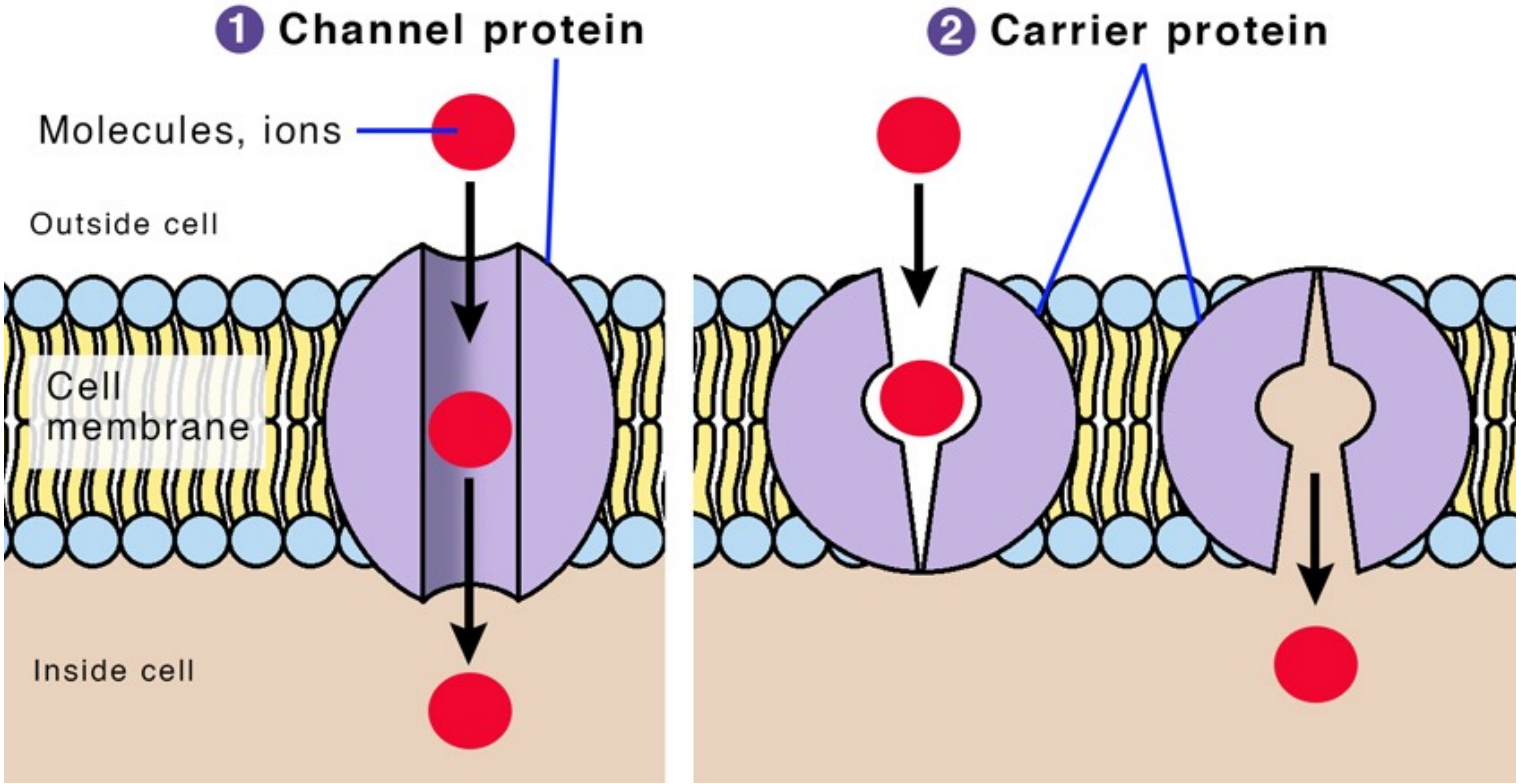
Energy Required

•Moves substances **against** their concentration or electrochemical gradient.
•**Requires energy**, usually in the form of **ATP** or an **ion gradient**.
•Mediated by **carrier proteins** (e.g., pumps, symporters, antiporters).
•Essential for maintaining **ion gradients**, **nutrient uptake**, and **cell volume**.
•Examples:

- **Na^+/K^+-ATPase** (primary active transport)
- **Na^+/glucose symporter** (secondary active transport using Na^+ gradient)
- **H^+-ATPase** in stomach lining

Carrier and Channel Proteins

- **Carrier proteins** bind specific molecules and undergo conformational changes.
- **Channel proteins** form pores allowing selective movement of ions/molecules.



Mechanism of Carrier Proteins

1.Substrate Binding
The carrier protein binds to a specific molecule (e.g., glucose or an amino acid) on one side of the membrane.

2.Conformational Change
Binding induces a change in the protein's shape, exposing the molecule to the other side of the membrane.

3.Molecule Release
The substrate is released into the intracellular or extracellular space, depending on the direction of transport.

4.Resetting
The carrier returns to its original conformation to repeat the process.

5.Types of Transport

1. **Facilitated diffusion**: No energy, down the concentration gradient.
2. **Primary active transport**: Uses ATP (e.g., Na^+/K^+-ATPase).
3. **Secondary active transport**: doesn't directly use ATP, but relies on the energy stored in the gradient established by the primary active transport. (e.g., Coupled transport: Na^+/glucose symporter, Calcium/hydrogen antiporter.

Mechanism of Channel Proteins

1.Pore Formation
The protein forms a **hydrophilic pore** across the membrane.

2.Selective Permeability
The channel is selective for specific ions or molecules (based on size and charge).

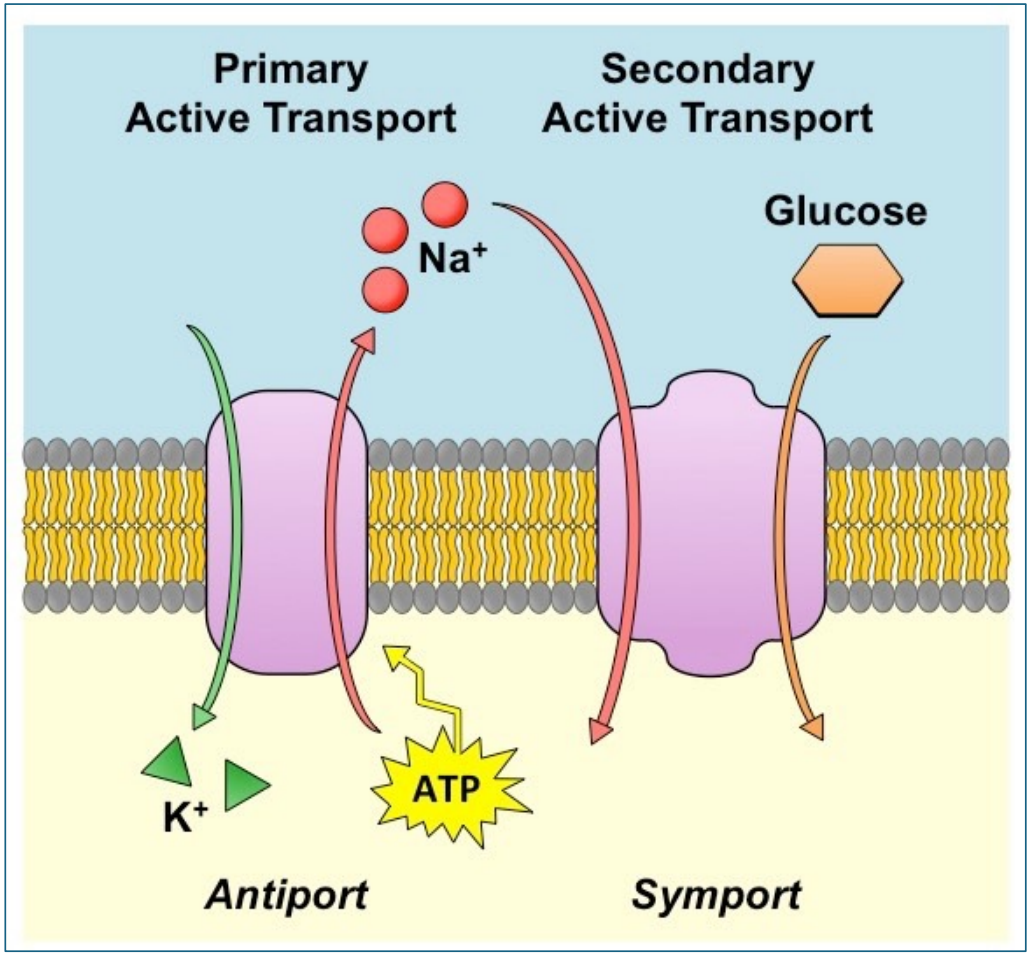
3.Opening and Closing (Gating)
Channels may be:

1. **Voltage-gated** (respond to membrane potential)
2. **Ligand-gated** (respond to chemical signals or ligands)
3. **Mechanically-gated** (respond to pressure or stretch)

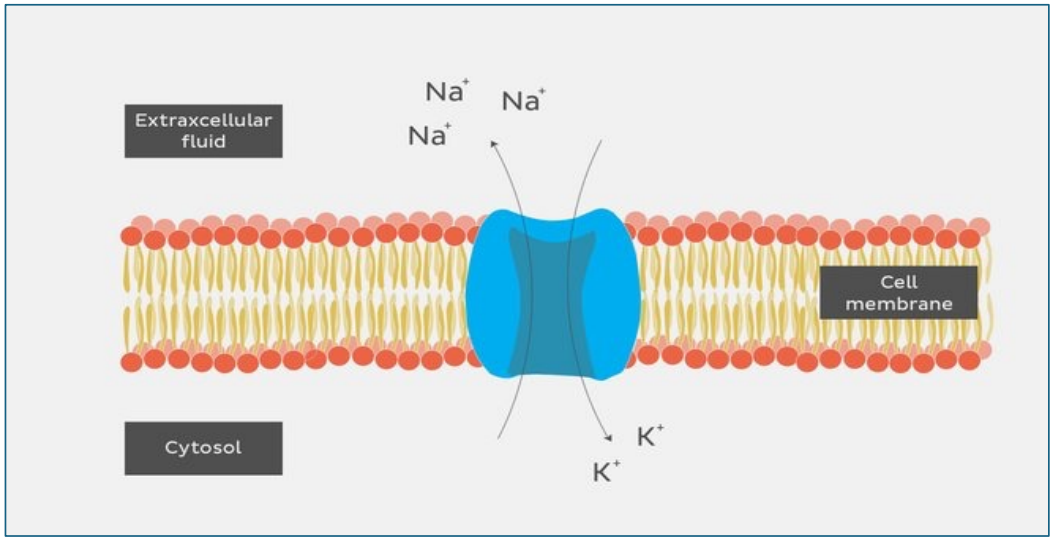
4.Passive Transport
Ions move **down their electrochemical gradient**, without the need for energy.

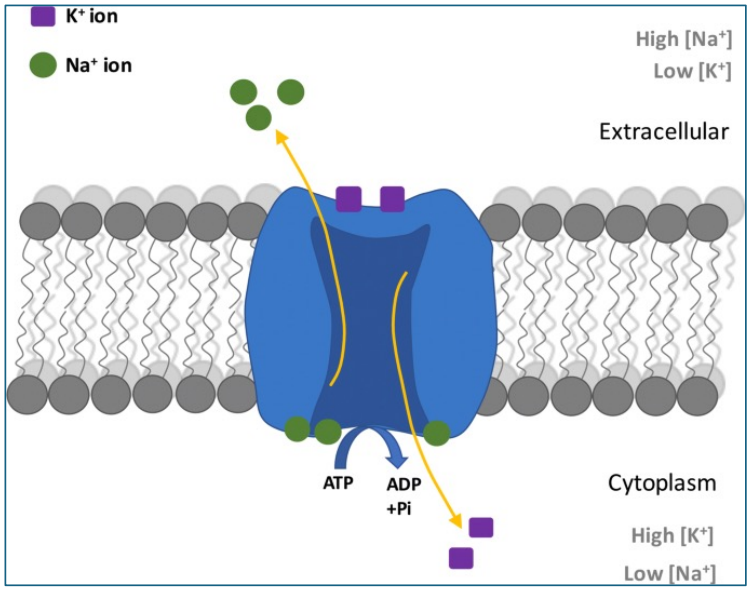
5.High Transport Rate
Channels allow rapid movement of thousands of ions per second, often much faster than carrier proteins.

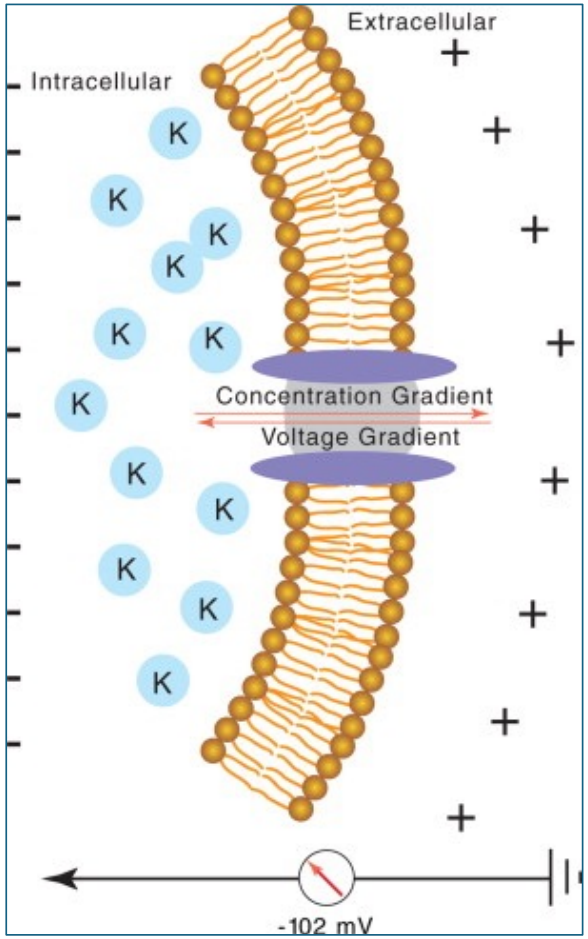
Mechanism of Carrier Proteins



Membrane Potential







CHANNEL PROTEINS VERSUS CARRIER PROTEINS

Channel proteins transport ions	Carrier proteins transport molecules
Position is fixed	Flip between two conformations
Solute molecules diffuse through the pores	Solute molecules are bound to one side and released from the other side
Have high transport rates	Have low transport rates
Do not bind with solute molecules it transports	Consist of alternative solute-bound conformations
Lipoproteins	Glycoproteins
Synthesized in the rough endoplasmic reticulum	Synthesized in the free ribosomes in the cytoplasm
Only transport water-soluble molecules	Transport both water soluble and insoluble molecules

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ACTIVE TRANSPORT VERSUS PASSIVE TRANSPORT

Active transport pumps molecules through the cell membrane against the concentration gradient	Passive transport allows molecules to pass the cell membrane through a concentration gradient
Utilizes cellular energy in the form of ATP	Does not require cellular energy
Endocytosis, exocytosis, secretion of substances into bloodstream and sodium/potassium pump are types of active transport	Diffusion, facilitated diffusion and osmosis are the types of passive transport
Allows molecules to pass the cell membrane, disrupting the equilibrium established by the diffusion	A dynamic equilibrium of water, nutrients, gasses and wastes is maintained between cytosol and extracellular environment
Ions, large proteins, complex sugars and cells are transported	Water soluble molecules like small monosaccharides, lipids, carbon dioxide, oxygen, and water are transported
Required for the entrance of large, insoluble molecules into the cell	Allows the maintenance of a delicate homeostasis between the cytosol and extracellular fluid

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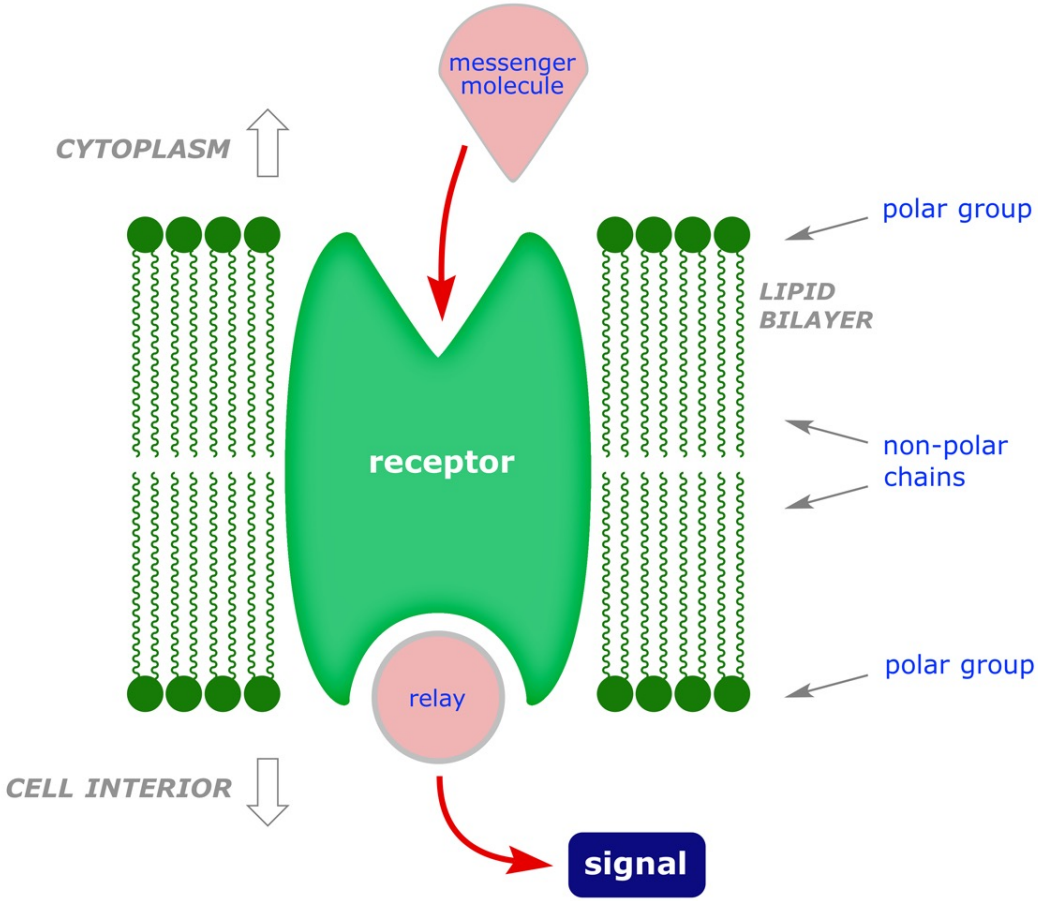
Protein as Receptors

- Receptors are **proteins that bind specific signaling molecules** (ligands) and trigger cellular responses.
- Each receptor recognizes and binds **specific ligands** (e.g., hormones, neurotransmitters, cytokines) with high affinity and selectivity.
- **Membrane receptors:** Located on the cell surface; respond to extracellular signals.
- **Intracellular receptors:** Found in cytoplasm or nucleus; respond to hydrophobic ligands (e.g., steroid hormones).
- Binding of a ligand activates a **signal transduction pathway**, converting an external signal into a cellular response.
- Receptors regulate **growth, metabolism, immune responses, neuronal communication**, and **homeostasis**.
- Ligand binding is typically **reversible**,



Types of Receptors

Cell Surface Receptors

- Located on the **plasma membrane** of the cell.
- Bind to **hydrophilic (water-soluble) ligands** that cannot cross the membrane (e.g., peptides, neurotransmitters).
- Trigger **signal transduction cascades** inside the cell (e.g., second messengers like cAMP, IP_3).
- Responses are typically **fast and short-term** (e.g., enzyme activation, ion flow).
- Examples:
  - **GPCRs** (e.g., β-adrenergic receptor)
  - **Receptor Tyrosine Kinases** (e.g., insulin receptor)
  - **Ion Channel Receptors** (e.g., nicotinic acetylcholine receptor)

Internal (Intracellular) Receptors

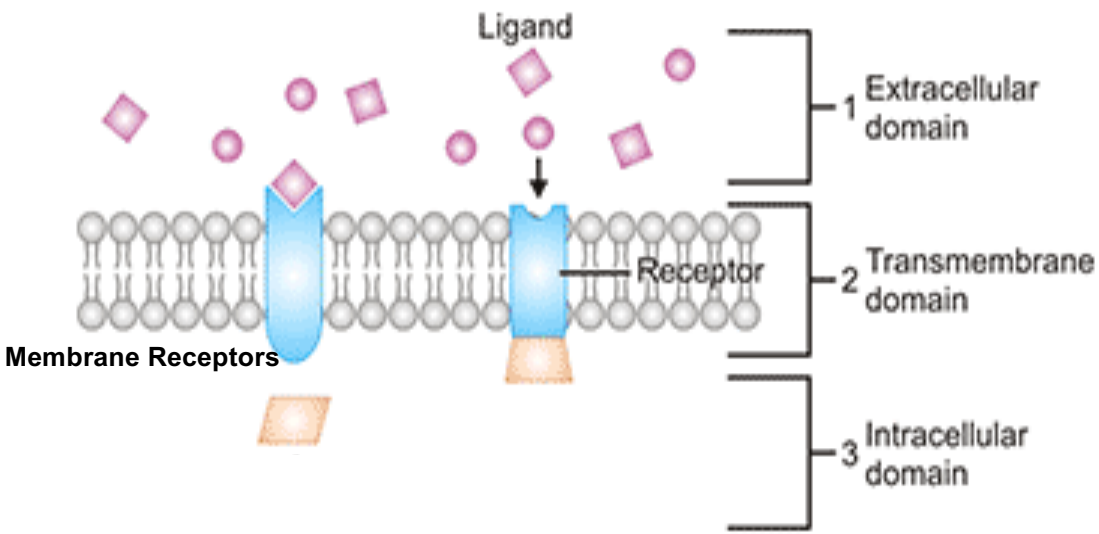
- Found in the **cytoplasm or nucleus** of the cell.
- Bind to **hydrophobic (lipid-soluble) ligands** that can diffuse through the membrane (e.g., steroid hormones).
- Function as **transcription factors**, directly regulating gene expression.
- Responses are typically **slower but long-lasting** (e.g., changes in protein synthesis).
- Examples:
  - **Glucocorticoid receptor**
  - **Estrogen receptor**
  - **Thyroid hormone receptor**

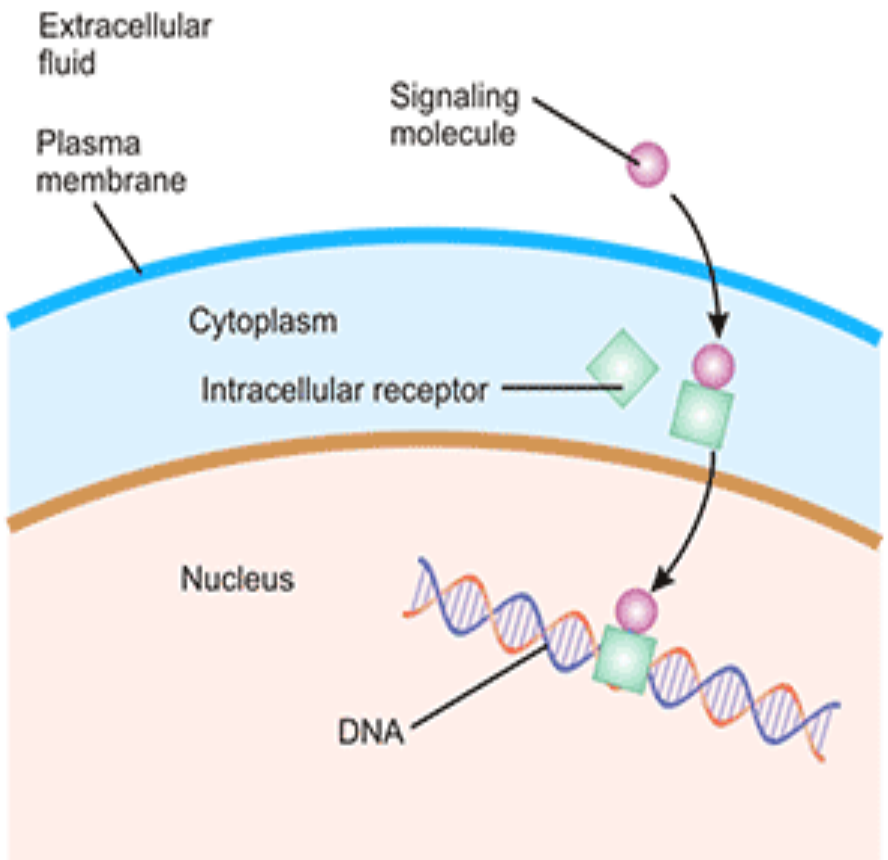
Membrane Receptors (extracellular ligands)

- G-Protein Coupled Receptors (GPCRs)
- Receptor Tyrosine Kinases (RTKs)
- Ion Channel Receptors

Intracellular Receptors (lipid-soluble ligands)

- Steroid hormone receptors (e.g., estrogen receptor)
- Thyroid hormone receptors





Protein as Structural elements

🔹 Structural proteins are specialized macromolecules that provide mechanical support, rigidity, and elasticity to cells and tissues.

🔹 They help maintain the three-dimensional architecture of biological systems.

🔹 **Localization:**

- **Intracellular Structural Proteins:** Found inside cells, especially as part of the cytoskeleton (e.g., actin, tubulin).
- **Extracellular Structural Proteins:** Located in the extracellular matrix (ECM) and connective tissues (e.g., collagen, elastin).

🔹 **Key Functions:**

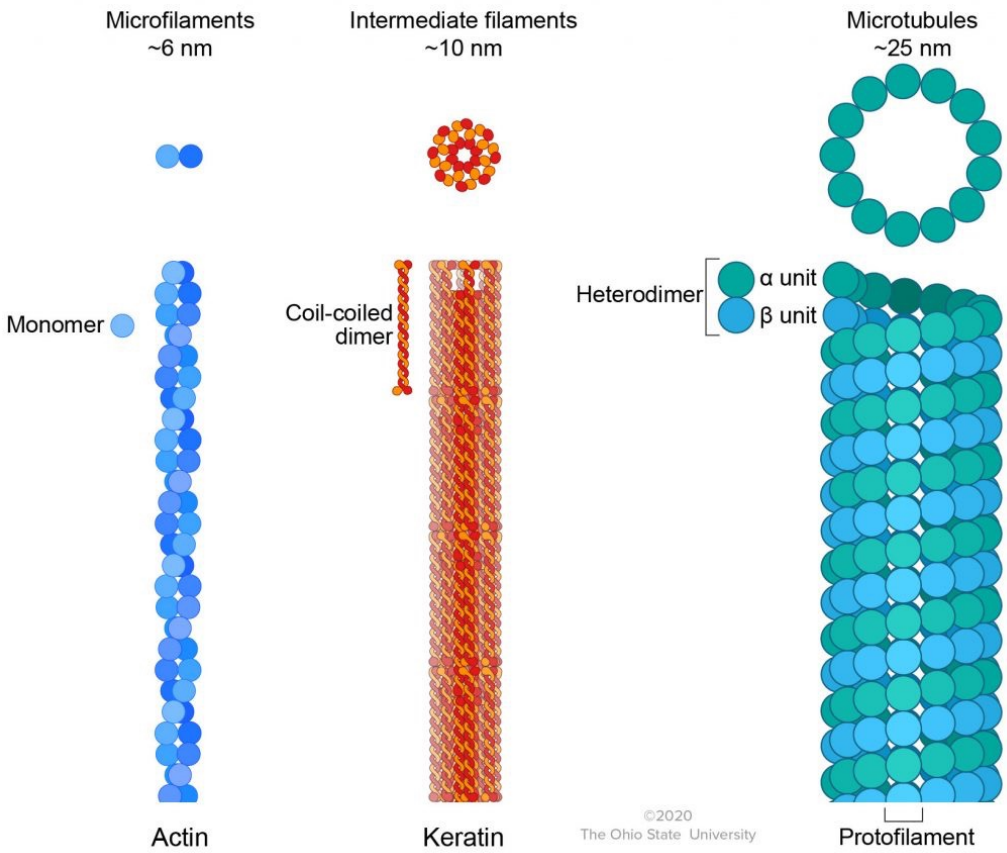
- Maintain cell shape and polarity.
- Support tissue integrity and organ structure.
- Anchor organelles and guide intracellular transport.
- Facilitate cell adhesion, migration, and signal transduction.

🔹 **Biological Importance:** Essential for development, wound healing, and immune cell mobility. Structural abnormalities are implicated in connective tissue disorders, skin diseases, and fibrotic conditions.

Cytoskeletal Structural Proteins

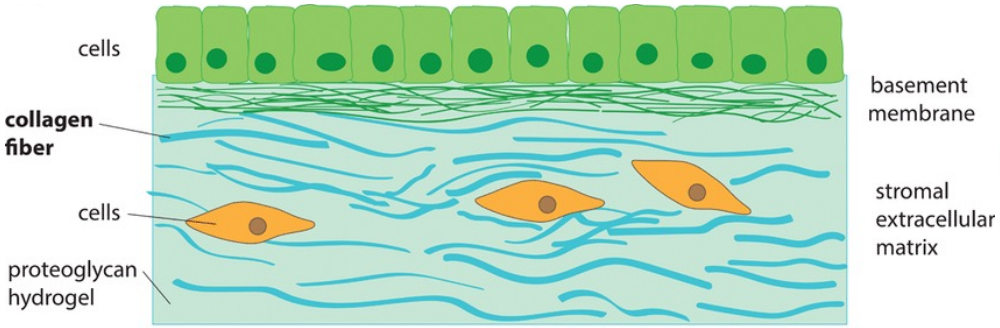
The cytoskeleton is a dynamic network of protein filaments that provides structural support, maintains cell shape, and facilitates intracellular transport and motility. It is composed of three main types of protein filaments: actin filaments, microtubules, and intermediate filaments.

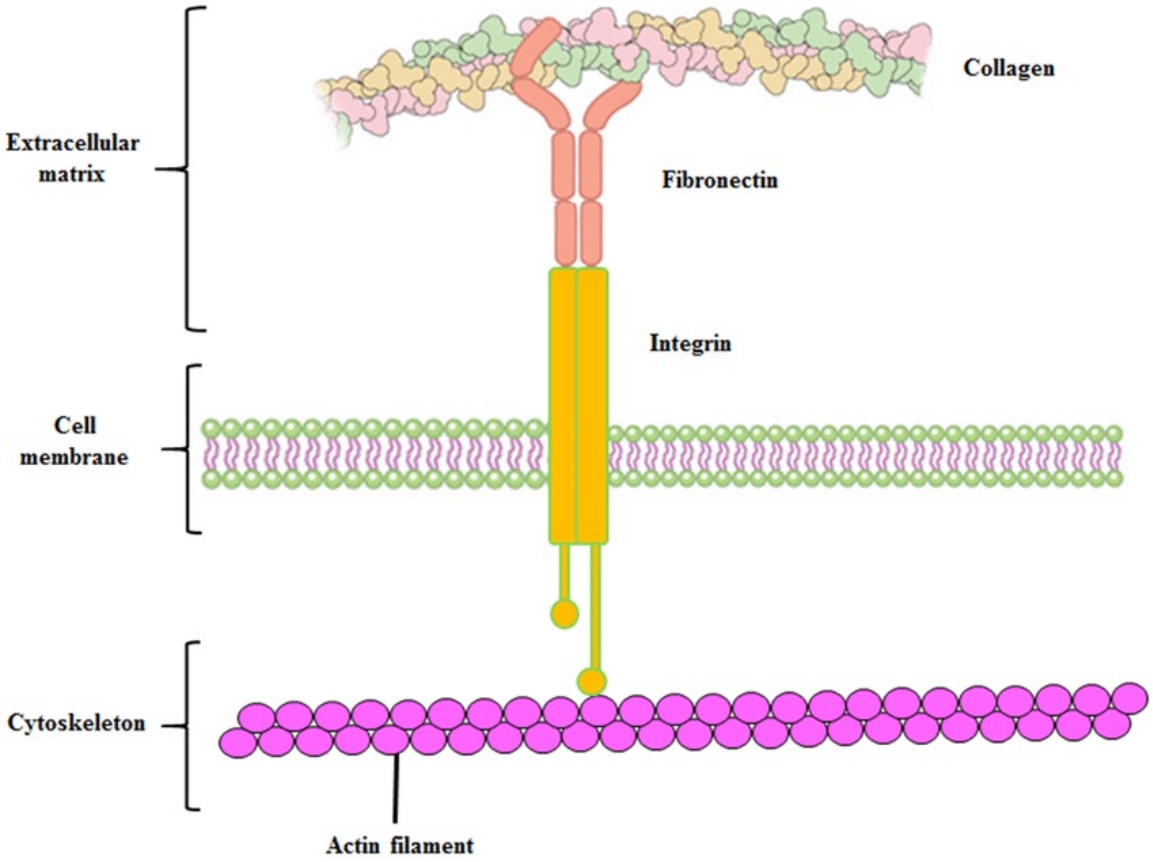
- **Actin Filaments (Microfilaments)**:
  - Composed of globular actin (G-actin).
  - Involved in **cell shape, motility, and cytokinesis**.
  - Forms structures like **lamellipodia** and **stress fibers**.
- **Microtubules**:
  - Made of α- and β-tubulin dimers.
  - Responsible for **intracellular transport**, **mitotic spindle formation**, and **cell polarization**.
- **Intermediate Filaments**:
  - Tissue-specific proteins like **keratins (epithelia)**, **vimentin (mesenchyme)**, and **neurofilaments (neurons)**.
  - Provide **mechanical strength and resilience** under stress.

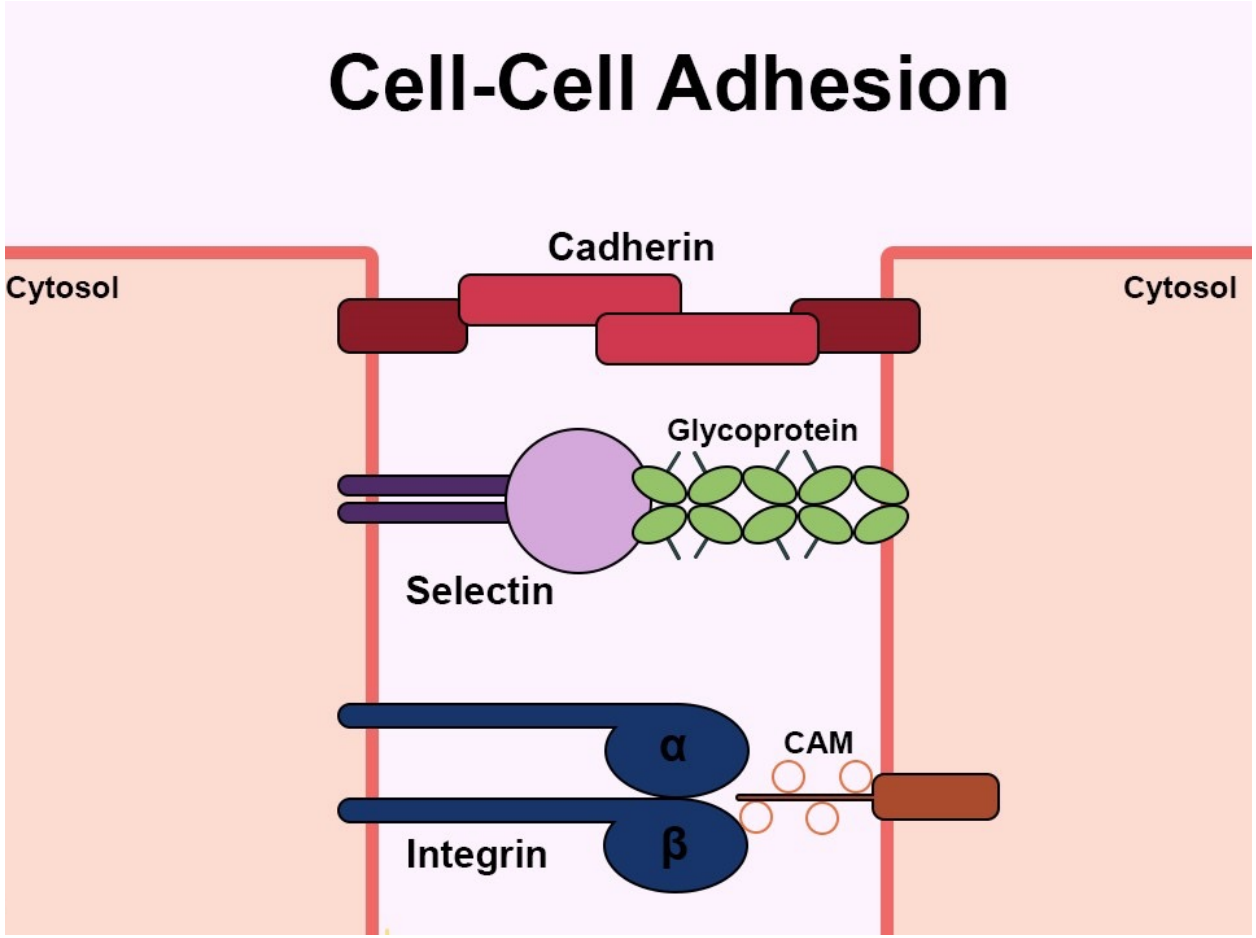


Structural Proteins in the Extracellular Matrix (ECM)

- **Collagen:** Major structural protein of ECM. Triple-helical structure; provides tensile strength in skin, bone, tendons.
- **Elastin:** Provides elastic recoil in tissues like lungs and arteries. Organized into elastic fibers interwoven with collagen.
- **Fibronectin:** Multi-domain protein aiding in cell adhesion, migration, and wound healing.
- **Laminin:** Major component of the basement membrane; involved in cell differentiation and tissue architecture.





Specialized Structural Proteins

- **Keratins:** Tough, fibrous proteins forming intermediate filaments in epithelial cells. Critical for the integrity of skin, hair, and nails.
- **Scaffold Proteins:** Provide a structural framework for signaling complexes (e.g., PSD-95 in neurons).Facilitate signal specificity and localization.
- **Cell Junction Structural Proteins:**
  - **Cadherins:** Mediate calcium-dependent cell–cell adhesion.
  - **Integrins:** Link ECM to cytoskeleton and participate in mechanotransduction.
  - **Selectins:** Selectins bind to sugar moieties and so are considered to be a type of lectin, cell adhesion proteins that bind sugar polymers.